IT532 & IT533: Introduction to Networks/Virtual Labs - SiteMap

This document can be used to help you follow along in the course. Check the boxes as you complete the pages.

Definitions
Discussion: These discussions provide opportunities for students to learn from each other. **Pre-assessment :** A test taken by learners before a new competency. To evaluate their knowledge & skills. 1 attempt per learner, per pre-assessment. Once students complete and submit the **Pre-assessment**, this **triggers the system** to unlock the **Formative Lessons**, **Milestone Activity**, and **Summative assessment**, creating a personalized pathway based on the score reflecting their prior knowledge. **Formative lessons** and **Formative assessments** provide opportunities to practice key concepts, allow you to check your answers, and prepare for the graded **milestone activities** and **summative assessments**. **Milestone Activity:** Student Mastery Required. The mastery (80% score) of the milestone assessment, allows students to move to a higher-level competency. **Summative Assessment:** Student Mastery Required. The mastery (80% score) of the summative assessment, allows students to move to a higher-level competency. Once you have mastered each competency by meeting the above requirements, your instructor will mark your course as **complete** with a letter "**P**" (**Pass**). This status can be viewed in the **gradebook** under "**Final Status**." **Program Continuation:** After completing IT532 & IT533, you are now ready to take the CompTIA Certification Network+ Exam.

Calbright Canvas Page CompTIA External Site ***Formative Lesson Header**

Virtual Lab Navigate to the labs section of the Network Topologies labs.

IT 532 Network Technology Click the button below to navigate back to Network Technology.

Orientations

- [x] Skillways Learner Orientation

Introduction

- [x] Welcome to the Calbright Network Technology Program
- [x] Meet Your Instructors
- [x] Network Technology Design
- [x] How This Course Works
- [x] Data Jobs in California
- [x] What is CompTIA
- [x] Getting Started

1.0: Open Systems Interconnection (OSI) Model (IT532 Introduction to Networks)

- [x] Welcome to Competency 1: Open Systems Interconnection (OSI) Model
- [x] 1.0: Discussion | Introduce Yourself
- [x] 1.0: Pre-Assessment

Formative Lesson 1.0

- [x] OSI Model Concepts
- [x] SOHO Networks
- [x] Internet Protocol Basics
- [x] Cloud Networking
- [x] 1.0: Formative Assessment Lesson

Milestone & Summative Assessments

- [x] 1.0: Milestone Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [x] 1.0: Summative Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [x] 1.0: Navigation to Network Technology Labs Virtual Lab Competency 10

10.0: Introduction to Network Virtual Lab (IT533 Introduction to Networks Virtual Lab)

- [x] Welcome
- [x] Introduction to PowerShell
- [x] Learn Git & GitHub
- [ ] GitHub Blog / Network Technology Portfolio
- [ ] Instructions for Packet Tracer Course (SkillsForAll)

- [ ] Cisco Packet Tracer Installation and Access Instructions for Lab Completion
- [ ] Packet Tracer Assignment Expectations/Submission Instruction

Exam and Interview Preparation

- [ ] Welcome

Network+ Exam and Interview Prep

- [ ] Interview Prep
- [ ] Preparing for the Network+ Certification Exam
- [ ] How to Breakdown Questions
- [ ] Taking the Network+ Assessment
- [ ] 10.0: Network Technologies Practice Exam
- [ ] 10.0: Navigation to Network Technology IT 532 Network Technology Network Technology Competency 2

2.0: Network Systems (IT532 Introduction to Networks)

- [x] Welcome to Competency 2: Network Systems
- [x] 2.0: Discussion | Network Systems
- [x] 2.0: Pre-Assessment

Formative Lesson 2.0A

- [ ] Routing Technologies
- [ ] Dynamic Routing Technologies
- [ ] Virtual LANs
- [ ] Disaster Recovery and High Availability
- [ ] Wireless Concepts and Standards
- [ ] Enterprise Wireless Network Design
- [ ] 2.0: Formative Assessment Lesson A

Formative Lesson 2.0B

- [ ] Wireless Security
- [ ] 2.0: Formative Assessment Lesson B

Formative Lesson 2.0C

- [ ] Virtual Private Networks
- [ ] Remote Management
- [ ] 2.0: Formative Assessment Lesson C

Milestone & Summative Assessments

- [ ] 2.0: Milestone Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [ ] 2.0: Summative Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [ ] 2.0: Navigation to Network Technology Labs Virtual Lab Competency 11

11.0: Network Topologies (IT533 Introduction to Networks Virtual Lab)

- [ ] Welcome-Network Topologies
- [ ] Lab I Office Network Design and Implementation
- [ ] Lab I Setting Up a Guest Wireless Network
- [ ] Lab I Classroom Debate on Wired vs. Wireless Networks
- [ ] Lab I Configuring Static Routing
- [ ] Lab I Peer Teaching on IP Address Management
- [ ] Lab I Configuring Quality of Service (QoS) for VoIP Traffic
- [ ] Lab I Configuring VLANs for Guest and Corporate Networks
- [ ] Lab I Configuring a Radius Server for Wireless Authentication

Milestone & Summative Assessments

- [ ] 11.0: Milestone Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [ ] 11.0: Summative Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [ ] 11.0: Navigation to Network Technology IT 532 Network Technology Network Technology Competency 3

3.0: IP Address Subclasses (IT532 Introduction to Networks)

- [ ] Welcome to Competency 3: Address Subclasses
- [ ] 3.0: Discussion | IP Address Subclasses
- [ ] 3.0: Pre-Assessment

Formative Lesson 3.0A

- [ ] Wiring Implementation
- [ ] Fiber Optic Cables and Connectors
- [ ] Physical Installation Factors
- [ ] 3.0: Formative Assessment Lesson A

Formative Lesson 3.0B

- [ ] Host Discovery and Monitoring
- [ ] Simple Network Management Protocol
- [ ] Packet Capture and Analysis

- [ ] Traffic Monitoring
- [ ] Switch Security
- [ ] 3.0: Formative Assessment Lesson B

Formative Lesson 3.0C

- [ ] Internet Protocol Basics
- [ ] IP Version 6
- [ ] Transport and Application Layer Protocols
- [ ] Dynamic Host Configuration Protocol
- [ ] Web, File, Print, and Database Services
- [ ] Email and Voice Services
- [ ] Simple Network Management Protocol
- [ ] Authorization and Account Management
- [ ] Remote Management
- [ ] 3.0: Formative Assessment Lesson C

Milestone & Summative Assessments

- [ ] 3.0: Milestone Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [ ] 3.0: Summative Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [ ] 3.0: Navigation to Network Technology Labs Virtual Lab Competency 12

12.0: Cabling (IT533 Introduction to Networks Virtual Lab)

- [ ] Welcome-Network Cabling
- [ ] Lab I Diagram Labeling Activity
- [ ] Lab I Implementing Power over Ethernet (PoE)y

Milestone & Summative Assessments

- [ ] 12.0: Milestone Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [ ] 12.0: Summative Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [ ] 12.0: Navigation to Network Technology IT 532 Network Technology Network Technology Competency 4

4.0: Network Addressing Schemes (IT532 Introduction to Networks)

- [ ] Welcome to Competency 4: Network Addressing Schemes
- [ ] 4.0: Discussion | Network Addressing Schemes
- [ ] 4.0: Pre-Assessment

Formative Lesson 4.0A

- ☐ Cable Troubleshooting
- ☐ Switch Troubleshooting
- ☐ IP Troubleshooting Tools
- ☐ Routing Technologies
- ☐ Virtual LANs
- ☐ Routing and VLAN Troubleshooting
- ☐ Transport and Application Layer Protocols
- ☐ DHCP Relay and Troubleshooting
- ☐ Domain Name System
- ☐ DNS Troubleshooting
- ☐ Web, File, Print, and Database Services
- ☐ Host Discovery and Monitoring
- ☐ Packet Capture and Analysis
- ☐ Spoofing Attacks
- ☐ Network Hardening
- ☐ Wireless Troubleshooting
- ☐ 4.0: Formative Assessment Lesson A

Formative Lesson 4.0B

- ☐ Networking Overview
- ☐ Enterprise Network Topologies
- ☐ Datacenter and Storage Networks
- ☐ 4.0: Formative Assessment Lesson B

Formative Lesson 4.0C

- ☐ IP Version 4 Addressing
- ☐ IP Version 4 Subnetting
- ☐ IP Version 6
- ☐ Routing Technologies
- ☐ APIPA and SLAAC
- ☐ 4.0: Formative Assessment Lesson C

Formative Lesson 4.0D

- [ ] Dynamic Host Configuration Protocol
- [ ] APIPA and SLAAC
- [ ] Domain Name System
- [ ] Application Security and Time Synchronization
- [ ] 4.0: Formative Assessment Lesson D

Milestone & Summative Assessments

- [ ] 4.0: Milestone Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [ ] 4.0: Summative Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [ ] 4.0: Navigation to Network Technology Labs Virtual Lab Competency 13

13.0: Network Addressing Schemes (IT533 Introduction to Networks Virtual Lab)

- [ ] Welcome-Network Addressing Schemes
- [ ] Lab I Configuring VLANs for Network Segmentation
- [ ] Lab I Setting Up and Configuring a DHCP Server
- [ ] Lab I Configuring Dynamic Routing with OSPF
- [ ] Lab I Setting Up a High-Availability Cluster
- [ ] Lab I Configuring DHCP Failover for High Availability
- [ ] Lab I Implementing IPAM for IP Address Management

Milestone & Summative Assessments

- [ ] 13.0: Milestone Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [ ] 13.0: Summative Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [ ] 13.0: Navigation to Network Technology IT 532 Network Technology Network Technology Competency 5

5.0: Network Communications Protocols (IT532 Introduction to Networks)

- [ ] Welcome to Competency 5: Network Communications Protocols
- [ ] 5.0 Pre-Assessment

Formative Lesson 5.0 A

- [ ] Cloud Concepts
- [ ] Cloud Networking
- [ ] 5.0: Formative Assessment Lesson A

Formative Lesson 5.0 B

- [ ] Networking Overview

- ☐ SOHO Networks
- ☐ Ethernet Switches
- ☐ Routing Technologies
- ☐ Firewalls
- ☐ Domain Name System
- ☐ Web, File, Print, and Database Services
- ☐ Disaster Recovery and High Availability
- ☐ Network Security Rules
- ☐ Zone-based Security
- ☐ Enterprise Wireless Network Design
- ☐ Wireless Security
- ☐ Wireless Troubleshooting
- ☐ Datacenter and Storage Networks
- ☐ Cloud Concepts
- ☐ Cloud Networking
- ☐ 5.0: Formative Assessment Lesson B

Milestone & Summative Assessments

- ☐ 5.0 Milestone Assessment (SUBMISSION IS REQUIRED TO GRADING)
- ☐ 5.0: Summative Assessment (SUBMISSION IS REQUIRED TO GRADING)
- ☐ 5.0: Navigation to Network Technology Labs Virtual Lab Competency 14

14.0: Network Communications Protocols (IT533 Introduction to Networks Virtual Lab)

- ☐ Welcome-Network Communications Protocols
- ☐ Quick Review: Networking Protocols
- ☐ Lab I Configuring Wireless Network Security
- ☐ Lab I Configuring an Intrusion Detection System (IDS)
- ☐ Lab I Configuring a Proxy Server
- ☐ Lab I Configuring Remote Desktop Access

Welcome-Practice Exams

- ☐ Taking the Network+ Assessment
- ☐ 14.0: Network Technologies Practice Exam

Milestone & Summative Assessments

- [ ] 14.0: Milestone Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [ ] 14.0: Summative Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [ ] 14.0: Navigation to Network Technology Network Technology Competency 6

6.0: Basic Understanding of Enterprise Network Technologies (IT532 Introduction to Networks)

- [ ] Welcome to Competency 6
- [ ] 6.0 Pre-Assessment

Formative Lesson 6.0A

- [ ] Internet Protocol Basics
- [ ] IP Version 6
- [ ] Transport and Application Layer Protocols
- [ ] Dynamic Host Configuration Protocol
- [ ] APIPA and SLAAC
- [ ] Domain Name System
- [ ] 6.0: Formative Assessment Lesson A

Formative Lesson 6.0B

- [ ] Web, File, Print, and Database Services
- [ ] Email and Voice Services
- [ ] Simple Network Management Protocol
- [ ] Authorization and Account Management
- [ ] Remote Management
- [ ] 6.0: Formative Assessment Lesson B

Formative Lesson 6.0C

- [ ] Ethernet
- [ ] Copper Cables and Connectors
- [ ] Wiring Implementation
- [ ] Fiber Optic Cables and Connectors
- [ ] Cable Troubleshooting
- [ ] Network Interfaces
- [ ] Wireless Concepts and Standards
- [ ] WAN and Internet Connectivity

- [ ] Datacenter and Storage Networks
- [ ] 6.0: Formative Assessment Lesson C

Formative Lesson 6.0D

- [ ] Networking Overview
- [ ] Enterprise Network Topologies
- [ ] Datacenter and Storage Networks
- [ ] 6.0: Formative Assessment Lesson D

Milestone & Summative Assessments

- [ ] 6.0 Milestone Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [ ] 6.0 Summative Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [ ] 6.0: Navigation to Network Technology Labs Virtual Lab Competency 15

15.0: Routing Network Assessment (IT533 Introduction to Networks Virtual Lab)

- [ ] Welcome-Routing Network Traffic
- [ ] Lab I Setting Up a Redundant Network Link
- [ ] Lab I Configuring Network Address Translation (NAT)
- [ ] Lab I Implementing Port Security on a Switch
- [ ] Lab I Implementing Network Segmentation with Subnetting
- [ ] Lab I Implementing IPsec for Secure Site-to-Site Communication

Milestone & Summative Assessments

- [ ] 15.0: Milestone Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [ ] 15.0: Summative Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [ ] 15.0: Navigation to Network Technology IT 532 Network Technology Network Technology Competency 7

7.0: Basic Network Administration (IT532 Introduction to Networks)

- [ ] Welcome to Competency 7: Basic Network Administration
- [ ] 7.0: Discussion | Basic Network Administration
- [ ] 7.0 Pre-Assessment

Formative Lesson 7.0A

- [ ] Network Interfaces
- [ ] Authentication
- [ ] Authorization and Account Management

- [ ] Network Hardening
- [ ] Switch Security
- [ ] Network Security Rules
- [ ] 7.0: Formative Assessment Lesson A

Formative Lesson 7.0B

- [ ] Zone-based Security
- [ ] Disaster Recovery and High Availability
- [ ] 7.0: Formative Assessment Lesson B

Formative Lesson 7.0C

- [ ] OSI Model Concepts
- [ ] Troubleshooting Methodology
- [ ] 7.0: Formative Assessment Lesson C

Formative Lesson 7.0D

- [ ] Ethernet
- [ ] Traffic Monitoring
- [ ] Enterprise Wireless Network Design
- [ ] Wireless Troubleshooting
- [ ] 7.0: Formative Assessment Lesson D

Milestone & Summative Assessments

- [ ] 7.0 Milestone Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [ ] 7.0: Summative Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [ ] 7.0: Navigation to Network Technology Labs Virtual Lab Competency 16

16.0: Network Systems (IT533 Introduction to Networks Virtual Lab)

- [ ] Welcome-Network Systems
- [ ] Lab I Setting Up a VPN for Remote Access
- [ ] Lab I Implementing a Network Firewall
- [ ] Lab I Setting Up Network Time Protocol (NTP)
- [ ] Lab I Configuring a DMZ for Secure Network Architecture
- [ ] Lab I Setting Up a Windows Domain Controller
- [ ] Lab I Configuring Network Redundancy with HSRP

Milestone & Summative Assessments

- [ ] 16.0: Milestone Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [ ] 16.0: Summative Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [ ] 16.0: Navigation to Network Technology Network Technology Competency 8

8.0: Network Security Threats & Vulnerabilities (IT532 Introduction to Networks)

- [ ] Welcome to Competency 8: Network Security Threats & Vulnerabilities
- [ ] 8.0: Discussion | Network Security: Threats & Vulnerabilities
- [ ] 8.0 Pre-Assessment

Formative Lesson 8.0A

- [ ] Copper Cables and Connectors
- [ ] Fiber Optic Cables and Connectors
- [ ] Cable Troubleshooting
- [ ] Network Interfaces
- [ ] Switch Port Configuration
- [ ] Switch Troubleshooting
- [ ] Email and Voice Services
- [ ] Wireless Troubleshooting
- [ ] 8.0: Formative Assessment Lesson A

Formative Lesson 8.0B

- [ ] Switch Port Configuration
- [ ] Switch Troubleshooting
- [ ] IP Troubleshooting Tools
- [ ] IP Troubleshooting
- [ ] Routing and VLAN Troubleshooting
- [ ] Network Security Rules
- [ ] 8.0: Formative Assessment Lesson B

Formative Lesson 8.0C

- [ ] Network Threats and Attacks
- [ ] Spoofing Attacks
- [ ] Rogue System Attacks
- [ ] Social Engineering

- [ ] Wireless Security
- [ ] Wireless Troubleshooting
- [ ] 8.0: Formative Assessment Lesson C

Milestone & Summative Assessments

- [ ] 8.0 Milestone Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [ ] 8.0: Summative Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [ ] 8.0: Navigation to Network Technology Labs Virtual Lab Competency 17

17.0: Basic Network Administration (IT533 Introduction to Networks Virtual Lab)

- [ ] Welcome-Basic Network Administration
- [ ] Lab I Troubleshooting Network Performance Issues
- [ ] Lab I Network Cable Damage and Replacement
- [ ] Lab I Identifying and Replacing Faulty Network Hardware
- [ ] Lab I Troubleshooting Wireless Connectivity Issues
- [ ] Lab I Implementing Link Aggregation
- [ ] Lab I Implementing a Content Filtering Solution
- [ ] Lab I Setting Up Network Access Control (NAC)
- [ ] Lab I Implementing a Load Balancer for Web Traffic
- [ ] Lab I Implementing Data Encryption for Network Communication

Milestone & Summative Assessments

- [ ] 17.0: Milestone Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [ ] 17.0: Summative Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [ ] 17.0: Navigation to Network Technology IT 532 Network Technology Network Technology Competency 9

9.0: Network Security Best Practices (IT532 Introduction to Networks)

- [ ] Welcome to Competency 9: Network Security Best Practices
- [ ] 9.0: Discussion | Network Security: Best Practices
- [ ] 9.0: Pre-Assessment

Formative Lesson 9.0A

- [ ] Networking Overview
- [ ] SOHO Networks
- [ ] Routing Technologies

- [ ] Firewalls
- [ ] Domain Name System
- [ ] Web, File, Print, and Database Services
- [ ] Disaster Recovery and High Availability
- [ ] Traffic Monitoring
- [ ] Network Security Rules
- [ ] Zone-based Security
- [ ] Enterprise Wireless Network Design
- [ ] Wireless Security
- [ ] Wireless Troubleshooting
- [ ] Datacenter and Storage Networks
- [ ] Cloud Concepts
- [ ] Cloud Networking
- [ ] 9.0: Formative Assessment Lesson A

Formative Lesson 9.0B

- [ ] OSI Model Concepts
- [ ] IP Version 6
- [ ] Modern Network Environments
- [ ] 9.0: Formative Assessment Lesson B

Formative Lesson 9.0C

- [ ] Security Concepts
- [ ] Network Threats and Attacks
- [ ] Spoofing Attacks
- [ ] Authentication
- [ ] Authorization and Account Management
- [ ] Network Hardening
- [ ] Zone-based Security
- [ ] Internet of Things
- [ ] Physical Security
- [ ] Wireless Security
- [ ] 9.0: Formative Assessment Lesson C

Milestone & Summative Assessments

- [ ] 9.0 Milestone Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [ ] 9.0: Summative Assessment - Practice Exam (SUBMISSION IS REQUIRED TO GRADING)
- [ ] 9.0: Navigation to Network Technology Labs Virtual Lab Virtual Lab Competency

18.0: Network Monitoring (IT533 Introduction to Networks Virtual Lab)

- [ ] Welcome-Network Monitoring
- [ ] Lab I Troubleshooting IP Address Conflicts
- [ ] Lab I Setting Up a Network Monitoring System
- [ ] Lab I Setting Up a File Server with RAID Configuration
- [ ] Lab I Configuring SNMP for Network Device Management
- [ ] Lab I Configuring a Wireless Controller for Centralized Management

Welcome-Practice Exams

- [ ] Taking the Network+ Assessment
- [ ] 18.0: Network Technologies Practice Exam

Milestone & Summative Assessments

- [ ] 18.0: Milestone Assessment (SUBMISSION IS REQUIRED TO GRADING)
- [ ] 18.0: Summative Assessment (SUBMISSION IS REQUIRED TO GRADING)

Durable Skills - OPTIONAL (IT533 Introduction to Networks Virtual Lab)

Durable Skills: Critical Thinking & Problem Solving

- [ ] Critical Thinking
- [ ] Peer Discussion: Critical Thinking Standards
- [ ] Critical Thinking Formative Assessment

Problem Solving Steps

- [ ] Problem Solving Formative Assessment
- [ ] Critical Thinking and Problem Solving Summative Assessment

Durable Skills: Written Communication

- [ ] Business Writing
- [ ] Interactive Activity: Workplace Email
- [ ] Bias in Writing
- [ ] Persuasive Writing
- [ ] Tailoring Writing
- [ ] Written Communication Summative Assessment

Durable Skills: Integrity and Reliability

- [ ] Recognizing Work Responsibilities and Time Management
- [ ] Optional Interactive Activity: Scheduling a Big Project
- [ ] Recognizing Work Responsibilities and Time Management Formative Assessment
- [ ] Customer Relationships
- [ ] Optional Activity: Peer Discussion
- [ ] Customer Relationships Formative Assessment

Integrity in the Workplace

- [ ] Integrity in the Workplace Formative Assessment
- [ ] Workplace Culture and Norms
- [ ] Optional Interactive Activity: Dressing Professionally for Different Work Cultures
- [ ] Workplace Culture and Norms Formative Assessment
- [ ] Integrity and Reliability Summative Assessment

Course Survey

- [ ] CompTIA Certification Exam
- [ ] Network Technology (IT533) - Course Survey

Program Continuation

- [ ] CompTIA Certification Exam